



Unlocking Relationships: The Power of Association

Section 1: Learn

What is Association in Data?

Association refers to **the relationship between two or more variables** in a dataset.

It helps in understanding **how changes in one variable affect another**.

Why is Understanding Association Important?

- **Identifies Patterns** – Helps find **relationships between different factors**.
- **Improves Decision-Making** – Businesses use it to analyze **customer behavior**.
- **Supports Predictions** – Helps in **forecasting trends** based on existing data.

Types of Association in Data

1. **Positive Association** – When one variable **increases, the other also increases**.

Example: More study hours → Higher exam scores.

2. **Negative Association** – When one variable **increases, the other decreases**.

Example: More speed → Less travel time.

3. **No Association** – No relationship between two variables.

Example: Shoe size and intelligence.



How Do We Measure Association?

- **Correlation Coefficient (r)** – Measures **how strong the relationship is** (-1 to +1 scale).
- **Scatter Plots** – Graphs that show **data relationships visually**.
- **Market Basket Analysis** – Finds product relationships in sales (e.g., “People who buy milk also buy bread”).

Anecdote: How Amazon Uses Association to Boost Sales

Amazon’s recommendation system uses **association rules** to suggest products. For example, if a user buys a **mobile phone**, they might get recommendations for **mobile cases and headphones**.

This strategy **increases sales and improves user experience**.

Section 2: Practice

1. Identifying Associations in Data

Example: Study Hours vs. Exam Scores

Study Hours	Exam Score (%)
2	40
4	55
6	70
8	85
10	90

- **Observation:** More study hours **lead to better scores** → **Positive Association**.



2. Understanding Negative Association

Example: Speed vs. Travel Time

Speed (km/hr)	Travel Time (minutes)
40	60
50	50
60	40
70	35
80	30

- **Observation:** As speed increases, travel time decreases → **Negative Association.**
-

3. Applying Market Basket Analysis

Example: Shopping Trends

Custo mer	Purchased Items
A	Milk, Bread
B	Milk, Bread, Butter
C	Milk, Eggs
D	Bread, Butter
E	Milk, Bread, Eggs, Butter



- **Insight:** People who buy **Milk** also buy **Bread (80% of the time)**.
Stores can use this insight for better product placement.
-

Section 3: Know More

Frequently Asked Questions (FAQs)

1. What is the difference between Correlation and Causation?

- Correlation means two variables move together but one does not cause the other.
- Causation means one variable directly influences the other.

2. How do we know if two variables are strongly associated?

If the correlation coefficient (r) is close to $+1$ or -1 , the relationship is strong.

3. Can two variables be associated but not related?

Yes! A high correlation does not mean one variable causes the other to change.

4. What is an example of No Association?

The number of umbrellas sold and the stock market performance.

5. How is Association used in real life?

- Healthcare: Finding relationships between lifestyle and diseases.
 - Marketing: Analyzing customer buying patterns.
 - Sports: Understanding player performance trends.
-



Conclusion:

Understanding associations in data helps in finding relationships, making predictions, and improving decision-making.

By using correlation, scatter plots, and market basket analysis, businesses and researchers can unlock powerful insights.